

# DJ Pal

## A Generative Art System for Musical Element Separation and Remix Creation

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### ABSTRACT

DJ Pal is a generative musical system designed to take any song and transform it into a genre-stylized remix of the user's choice. The system takes a single audio file as input and applies a combination of digital signal processing (DSP) techniques and artificial intelligence models to produce a user-customized remix. DJ Pal makes use of stem separation through Hybrid Demucs to isolate vocals, drums, bass, and other instruments in the uploaded song. Users then select from the list of extracted stems along with a genre style to generate their remix of the song. Based on the genre chosen by the user, DJ Pal will alter each selected stem through tempo detection and adjustment, pitch shifting, equalization, and rhythm enhancements. Additionally, users can choose to add a genre-specific backing track produced using the transformer-based generative model, MusicGen.

DJ Pal gives users the ability to remix their favourite songs into a different genre, which typically would be hard for beginners to achieve. This is because most music is only available as a single mixed audio file. DJ Pal's ability to extract audio stems from any song while allowing users to handselect any of the stems to be remixed into a different genre allows for music to be explored in ways previously inaccessible.

### 1 Introduction

Remixing is a routine practice in music production, as it allows artists to reinterpret existing songs. However, traditional remixing practices require knowledge in audio engineering and access to expensive, specialized software. DJ Pal lowers this barrier by automating the most complicated and challenging parts of the remixing process, while still granting users control over the creative process. DJ Pal takes a single WAV file as input and then outputs a remixed version based on user selections. Users can choose the genre and what stems to remix. DJ Pal does not generate a new song from scratch, rather the system focuses on transforming existing songs. The identity of the original song is ensured to be preserved while new creative interpretations are explored by users.

DJ Pal, as a concept, is innovative and interesting because it combines digital signal processing techniques with modern AI. Hybrid Demucs is used to separate musical stems, while MusicGen provides generative musical tracks to optionally

complement the output. This combination makes up DJ Pal's hybrid of modern and older generative techniques.

The motivation behind this project is to explore artificial intelligence as a creative collaborator with humanity itself. DJ Pal gives humans a say in how they would like music to be remixed, while ensuring generative systems remove the requirement for expertise in musical remixing.

## 2 Background and Related Work

### 2.1 Audio Source Separation

Audio source separation is the process of isolating individual musical components (stems) from an audio mix (Manilow et al., 2020). Hybrid Demucs in particular achieves high-quality stem separation using a hybrid waveform and spectrogram-based architecture (Défossez, 2021). The use of Hybrid Demucs allows DJ Pal to isolate vocals, drums, bass, and other elements for transformation.

### 2.2 Digital Signal Processing in Music

Digital signal processing (DSP) grants DJ Pal the ability to analyze and transform audio through mathematics (Analog Devices, n.d.).

The Python library, Librosa, provides DSP tools for extracting tempo, pitch, and other spectral features. These extractions enable the use of time-stretching, pitch shifting, and harmonic-percussive separation through Librosa (McFee et al., 2015).

### 2.3 Generative Music Models

MusicGen is a transformer-based generative model that generates music from text-based prompts. It learns from a training set of musical structures such as tempo and genre (Copet et al., 2023). DJ Pal incorporates MusicGen for the purpose of generating optional backing tracks to enhance the remix's genre-stylized output.

## 2.4 Computational Creativity in Music

DJ Pal matches Margaret Boden's concept of exploratory creativity, which is one of Boden's three classifications of creativity. This is because DJ Pal explores the already established space of music and genre. The system works within the space to transform existing music through various musical concepts and rules (Boden, 2004). DJ Pal overall contributes to computational creativity through its combination of analyzing, transforming, and generating music through a single unique pipeline.

## 2.5 Comparison to Existing Systems

Before DJ Pal, existing AI musical systems typically focused on stem separation or music generation separately. For example, Hybrid Demucs separates music into individual stems (Défossez, 2021). While MusicGen generates exclusively new songs from text prompting (Copet et al., 2023). These systems are independent from each other, whereas DJ Pal combines them into one unified and user-friendly pipeline.

Remixing systems involving source separation do exist, but they lack rule-based genre transformations and generative music features. For example, there are end-to-end neural remixing pipelines that only let users adjust the strength of the stems (Yang et al., 2021).

Overall, DJ Pal is different from these systems due to its stem separation, rule-based signal processing, and generative AI single pipeline approach. DJ Pal is not purely generative due to the system's use of existing audio and its preservation of the original song structure. The system's allowance of users to choose what stems to apply genre transformations to, and to what genre, allows for user control not usually seen in a fully automated system.

# 3 System Design and Implementation

## 3.1 Overview

DJ Pal is an audio remixing system with many key steps in its single

pipeline:

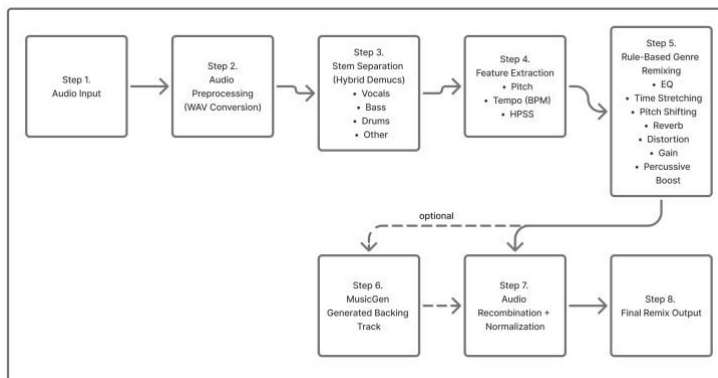


Figure 1: DJ Pal's pipeline diagram.

## 3.2 Audio Preprocessing

The user will input an audio file into DJ Pal. Once received by DJ Pal, it is converted to WAV format, if not already in WAV format. This step ensures compatibility with all processing steps in DJ Pal. This also helps avoid inconsistencies with the audio.

## 3.3 Stem Separation

DJ Pal uses Hybrid Demucs, a Python deep learning model for stem separation (Défossez et al., 2019). The inputted song is separated into four stems: vocals, bass, drums, and other uncategorized instruments. This is an important step for DJ Pal as it allows the genre transformations to be applied to specific user-selected stems rather than to the whole song itself.

## 3.4 Genre-Based Remixing

Users can choose between these genres for remixing:

- EDM
- Hip-Hop
- Lo-Fi
- Rock
- Industrial
- Punk
- Pop
- Metal
- Cinematic
- Grunge
- Dubstep
- Techno

Each genre in the list is defined by a set of rule-based parameters that keep genre stylizing consistent, which are:

- Tempo (BPM)
- Musical Key
- Equalization (EQ)
- Reverb
- Distortion
- Gain
- Percussive Boost

These parameters are transformed using DSP techniques:

- Time-stretching to match the pre-defined genre target tempo
- Pitch detection and shifting to match the pre-defined genre target key
- Genre-specific EQ adjustments
- Genre-specific reverb adjustments
- Genre-specific distortion adjustments
- Genre-specific gain adjustments

- Harmonic/Percussive separation for genre-specific percussive boosts

**3.4.1 Frequency Domain Equalization.** The equalization adjustment uses a function that implements the Discrete Fourier Transform (DFT) (NumPy, n.d.). This is how audio signals are transformed from the time domain to the frequency domain (Mintlify, n.d.). The Discrete Fourier Transform (DFT) equation is as follows (Delgutte & Greenberg, 1999):

$$X[k] = \sum_{n=0}^{N-1} x[n] e^{-j2\pi kn/N}$$

This process allows for DJ Pal to select what frequencies to boost or lower. For example, DJ Pal boosts low frequencies for the EDM genre and lowers high frequencies for the Lo-Fi genre.

**3.4.2 Time-Scale Modification.** DJ Pal's tempo matching is applied through Librosa time stretching. The time is stretched or compressed using the ratio created by the tempo multiplier equation (Librosa, n.d.):

$$\text{tempo multiplier} = \frac{\text{target BPM}}{\text{original BPM}}$$

**3.4.3 Pitch Shifting and Key Alignment.** Pitch shifting is used by DJ Pal to match a genre-specific target musical key. Pitch is shifted through semitone scaling to get to this target musical key (Librosa, n.d.). The number of steps to scale is calculated by using the index of the target key and the index of the original key:

$$n_{\text{steps}} = (\text{index}_t - \text{index}_o) \bmod 12$$

**3.4.4 Harmonic-Percussive Source Separation (HPSS).** DJ Pal makes use of Harmonic-Percussive source separation (HPSS). Harmonic and percussive components of the stem are separated using Librosa (Librosa, n.d.). Once the percussive component is separated, it is boosted by multiplying a genre multiplier by the percussive source. It is then added back together with the harmonic source:

$$y = y_h + (\text{boost}) y_p$$

## 3.5 Generative AI Backing Tracks

DJ Pal allows users to select whether they would like to incorporate an AI-generated backing track in their remix using MusicGen. DJ Pal has pre-defined text prompts describing each genre that will generate a short audio clip.

The generated MusicGen track is kept short to ensure DJ Pal runs as fast as possible. To match the length of the inputted song, DJ Pal loops the audio with a crossfade effect from Pydub for a less noticeable loop. Once the MusicGen track is looped, it is overlayed on the existing stems. MusicGen helps enhance the final remix without destroying the structure of the original input.

## 3.6 Audio Mixing and Normalization

Remixed and non-remixed stems are recombined in the final output. A loudness normalization is applied to prevent clipping/audio distortion and to ensure a consistent output.

## 3.7 Visualization and Analysis

The evaluation of the system is done through generated visual comparisons of the original song and the remixed version. The visuals outputted are:

- Spectrograms (Mel spectrogram)
- Rhythm (onset strength) plots (rhythmic intensity)
- Pitch (chroma) visualizations (pitch distribution)
- Energy (RMS) graphs (perceived loudness)

## 3.8 User Interface

The user interface is created using Gradio, a Python library that assists in creating interfaces for machine learning (Gradio, n.d.). Through the interface, users upload an audio file, preview the four stems that were separated, select their genre, stems to remix, and whether or not they want an AI-generated backing track, and then generate and download their final remix. The user interface is built to guide users step by step, intuitively, by starting with step one at the top of the interface.

## 3.9 Implementation Details

DJ Pal makes use of these Python libraries for its implementation:

- Librosa for audio analysis and transformations
- Hybrid Demucs for source separation
- Pydub for audio manipulation
- HuggingFace transformers for MusicGen's integration
- Gradio for the user interface
- Soundfile for saving and loading audio files
- Matplotlib for technical visualizations
- Scipy for the add reverb function convolution application
- Numpy for mathematical applications
- Os for file handling
- Torch and Torchaudio for working with audio data

## 4 Results and Evaluation

### 4.1 Qualitative Evaluation

The system is somewhat successful in its goal of producing genre-stylized remixes through rule-based DSP. However, genre transformations tend to be strongest when a MusicGen backing track is used.

For example:

- EDM remixes should have an increase in bass and tempo intensity

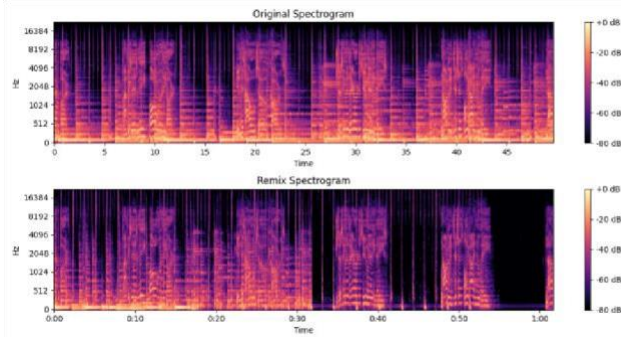


Figure 2: Spectrogram before and after of a country song. The genre chosen for DJ Pal was an EDM remix (no MusicGen used, all stems chosen for remix).

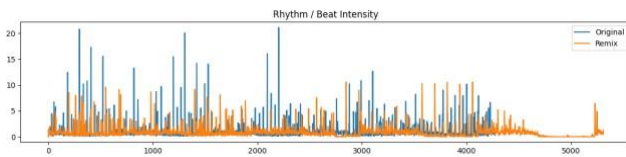


Figure 3: Rhythm before and after of a country song. The genre chosen for DJ Pal was an EDM remix (no MusicGen used, all stems chosen for remix).

Based on Figure 2, it appears that DJ Pal is not fully successful in implementing a higher amplitude for its EDM remixes. Figure 3 shows the remix to have less rhythm intensity than the original. This could potentially be due to normalization processes, but it is unclear.

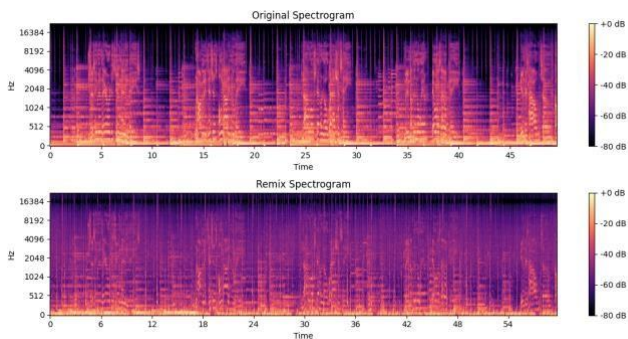


Figure 4: Spectrogram before and after of a country song. The genre chosen for DJ Pal was an EDM remix (MusicGen used, all stems chosen for remix).

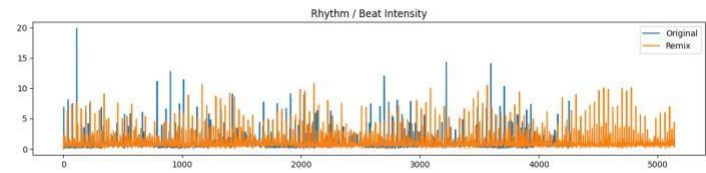


Figure 5: Rhythm before and after of a country song. The genre chosen for DJ Pal was an EDM remix (MusicGen used, all stems chosen for remix).

Based on Figure 4, it appears that DJ Pal is more successful in implementing a higher amplitude for its EDM remix. The overlaid MusicGen backing track seen in the spectrogram seems to help out with this. Figure 5 shows that when remixed with the added MusicGen backing track, there is an overall greater rhythm intensity than the original. This shows that MusicGen creates a more successful remix.

- Lo-Fi remixes should have low energy and soften any high frequencies.

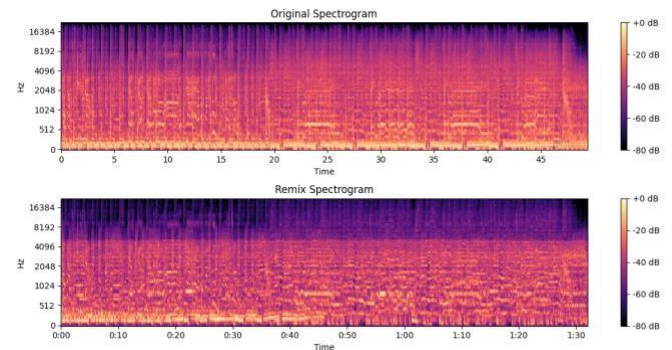


Figure 6: Spectrogram before and after of a rock song. The genre chosen for DJ Pal was a Lo-Fi remix (no MusicGen used, all stems chosen for remix).

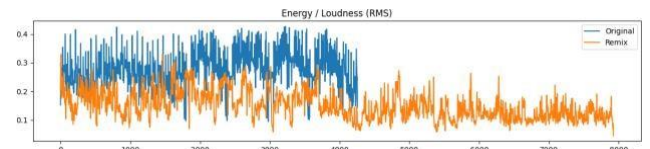


Figure 7: Energy before and after of a Rock song. The genre chosen for DJ Pal was a Lo-Fi remix (no MusicGen used, all stems chosen for remix).

Based on Figure 6, DJ Pal has succeeded in softening the high frequencies in the rock song. Figure 7 also shows that energy has been effectively lowered.

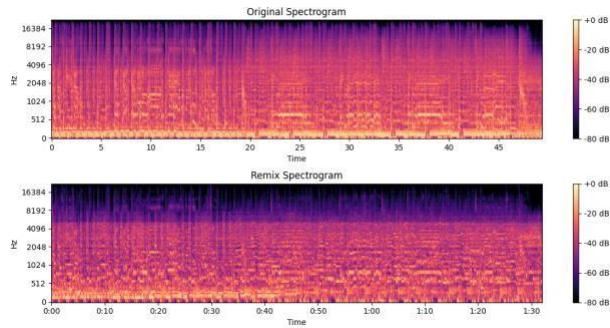


Figure 8: Spectrogram before and after of a rock song. The genre chosen for DJ Pal was a Lo-Fi remix (MusicGen used, all stems chosen for remix).

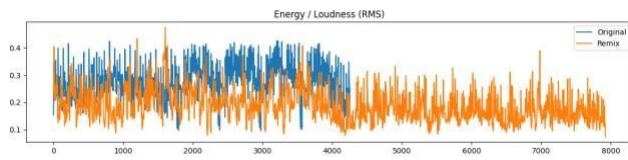


Figure 9: Energy before and after of a Rock song. The genre chosen for DJ Pal was a Lo-Fi remix (MusicGen used, all stems chosen for remix).

Based on Figure 8, DJ Pal has succeeded in softening the high frequencies in the rock song with MusicGen. Figure 9 also shows that energy has been effectively lowered with the MusicGen backing track included. In this example, both Lo-Fi remixes were successful.

Overall, these visualizations show that DJ Pal is somewhat effective in its remixing modifications. However, when MusicGen backing tracks are added, the strongest outputs are produced.

## 4.2 Limitations

Although DJ Pal is successful, it does have some limitations. The system tends to work best when transforming multiinstrumental songs; some of DJ Pal's transformations are not as meaningful to the final remix. For example, a pianoexclusive song will have limitations on how much it will transform to match its target genre. It should also be noted that in this case, every stem will be empty, except for the "other" stem. As well, another limitation is that the quality of stem separation depends on the audio users upload to DJ Pal.

In terms of the AI-generated backing track using MusicGen, the feature only creates short audio clips. Unfortunately, if the AI audio were lengthier, it would take too long to generate. DJ Pal is slow in creating remixes in general due to its implementation of two AI models.

Pitch shifting can have some issues in the vocal stem. Pitch shifting may create unnatural-sounding vocals. In

response to this issue, a limit on pitch shifting for vocals has been applied. However, it should be noted that this may not always fix the problem in an impactful enough way.

Finally, one last limitation is that DJ Pal is currently limited to only twelve genres. This is because of time constraints and the importance of refining the genres currently implemented.

## 5 Creative and Technical Implications

DJ Pal's creative implications are its demonstration in computational creativity, specifically Boden's exploratory creativity (Boden, 2004). DJ Pal takes the existing space of remixing and explores how it can be done uniquely, but with reference to traditional methods. Instead of creating entirely new content, DJ Pal focuses on transforming existing audio through various musical concepts. Generative AI music is still a feature in DJ Pal, but it is implemented so that it does not take away structure from the original song.

One thing to note is that DJ Pal does not aim to replace human creativity. DJ Pal allows anyone to explore remixing and can be used for quick prototypes or to inspire users to learn real remixing techniques themselves. DJ Pal is meant to give the accessible decisions of remixing to its human user, which are genre and stem selection. The system, however, handles the challenging technical and less accessible parts. DJ Pal opens new possibilities in music production accessibility and creativity beyond an individual's typical skill set.

## 6 Conclusion

DJ Pal is a multi-step generative musical remix system. It employs stem separation, digital signal processing, and a generative AI model to create meaningful outputs. The system reaches its goal in creating stylistically matched and recognizable genre-transformed remixes, but is strongest when a MusicGen backing track is used. The system ensures users continue to have creative control, but technical knowledge around remixing is not required for DJ Pal. DJ Pal is not only successful, but accessible as well. Overall, DJ Pal is a

unique system that combines several generative techniques into one working system.

### 6.1 Future Work

DJ Pal's future would involve improvements to create an even stronger system. For example, DJ Pal could make use of more advanced stem separation and generative music models. This could improve outputs, along with improving remixing speeds. Another

way DJ Pal could be improved is by adding more user controls, but in a way that ensures minimal confusion for those not well-versed in music theory and remixing. For example, DJ Pal could implement volume sliders for stems that would affect how loud each stem sounds in the final remix. Finally, along with adding more genres, the system could give users the ability to mix genres. For example, users could create a Pop Punk remix or an Industrial Metal remix.

## 7 Acknowledgments

DJ Pal makes use of several open-source tools, including Hybrid Demucs, Librosa, MusicGen, HuggingFace Transformers, and Gradio. Testing of DJ Pal was done using royalty-free music from YouTube's Creator Audio Library. DJ Pal was written in the Python language.

### 7.1 AI Use Acknowledgement

ChatGPT has been used for assistance in code debugging, generating initial MusicGen prompts and genre values to work with (reverb values, gain values, percussive boost values, target tempo values, and target key values), along with assistance in using and learning librosa. ChatGPT helped me create genre-specific if-else statements/values for further reverb enhancements and genre enhancements, create the genre EQ function and values, and create and use a distortion effect function. The add reverb function was created with Google Gemini.

## 8 Ethics Statement

Due to copyright concerns, DJ Pal's testing makes use of royalty-free music from YouTube's Creator Audio Library. It should be noted that DJ Pal's purpose is not to replace human musicians but rather to inspire human creativity in beginner remixers.

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